PRATIK PATIL

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Summary

Generative AI Engineer with 2+ years of professional experience designing and deploying production **Large Language Model (LLM)** solutions, holding an MSc in Artificial Intelligence. Currently at Silverspace, delivering **Retrieval Augmented Generation (RAG)** platforms, **agentic AI** and **multi-agent** workflows using **LangChain**, **LangGraph**, **Model Context Protocol (MCP)** and **vector search**, alongside **fine-tuned LLMs** and rigorous **LLM evaluation**. Experienced in deploying **FastAPI** services on **AWS** and **Azure** with **Docker**, **Kubernetes** and **CI/CD**. Prior AI engineering in robotics at vizva Consultancy Services spanning **Python**, **PyTorch**, **computer vision** and **MLOps**. Microsoft Azure AI Fundamentals (AI-900) certified.

Technical Skills

Generative AI & LLMs: Generative AI (GenAI), Large Language Models (LLMs), Retrieval Augmented Generation (RAG), Prompt Engineering, Fine-Tuning, LoRA, QLoRA, Hugging Face Transformers, OpenAI API, Llama, Mistral, Embeddings
Agentic AI & Orchestration: Agentic AI, Multi-Agent Systems, LangChain, LangGraph, LlamaIndex, Model Context Protocol (MCP), Tool Calling, Function Calling, AI Guardrails, LLM Evaluation, RAGAS, Hallucination Mitigation
Vector Search & Data: Vector Databases, Pinecone, FAISS, ChromaDB, pgvector, Semantic Search, Reranking, Chunking Strategies, PostgreSQL, SQL, ETL Pipelines, Data Analytics
Languages & Frameworks: Python, SQL, FastAPI, Flask, REST APIs, PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, Pydantic, Git
Cloud, MLOps & Deployment: Amazon Web Services (AWS), Microsoft Azure, Azure AI Services, Docker, Kubernetes (K8s), CI/CD, GitHub Actions, MLflow, LLMOps, Model Quantization, Model Distillation, Observability
Machine Learning & Robotics: Machine Learning (ML), Deep Learning, Natural Language Processing (NLP), Neural Networks, Computer Vision, OpenCV, Reinforcement Learning, Robot Operating System (ROS)

Experience

Silverspace Remote
Generative AI Engineer *Nov 2025 – Present*

- Architected **Retrieval Augmented Generation (RAG)** pipelines with **LangChain**, **LlamaIndex**, **embeddings**, **reranking** and **vector databases** over 8,000 documents, giving 150 users cited answers.
- Engineered **multi-agent agentic AI** workflows with **LangGraph**, **tool calling** and **Model Context Protocol (MCP)** servers, automating 3 multi-step business workflows and saving roughly 40 staff hours per month.
- Fine-tuned **Large Language Models (LLMs)** such as **Llama** and **Mistral** with **LoRA**, **QLoRA** and **Hugging Face Transformers** on 12,000 examples, cutting inference spend by £1,500 monthly.
- Deployed containerised **FastAPI** inference services with **Docker**, **Kubernetes (K8s)** and **CI/CD** on **Amazon Web Services (AWS)**, serving 5,000 daily **LLM** requests with response times under 2 seconds.
- Instituted **LLM evaluation** with **RAGAS**, prompt regression tests and **guardrails** across 250 test cases, catching 60 **hallucination** and prompt injection failures before 4 production releases.
- Streamlined **prompt engineering** and **LLMOps** practice through versioned prompts, **MLflow** experiment tracking, shortening **Generative AI** release preparation from 3 days to 1 day.

vizva Consultancy Services India
Junior AI Engineer (Robotics) *Feb 2023 – Apr 2024*

- Developed **computer vision** and **reinforcement learning** pipelines in **Python**, **PyTorch** and **OpenCV** on 15,000 images for pick-and-place via **Robot Operating System (ROS)**, saving 4 seconds per part.
- Automated sensor-data preprocessing, model validation and reporting with **Python**, **Pandas** and **NumPy**, saving roughly 120 engineering hours over 14 months of **machine learning** training cycles.
- Prototyped **deep learning** and **neural network** models for object detection and motion planning, delivering 3 working demonstrations to client stakeholders within 10 weeks.

Projects

AssistFlow AI – Agentic Customer Support Platform | *LangGraph, RAG, Tool Calling, FastAPI* MSc Project

- Orchestrated **multi-agent agentic AI** support flows covering triage, **RAG**-based answer generation and human escalation, with **tool calling** into business systems for end-to-end ticket resolution.
- Integrated **guardrails**, conversation memory and **LLM evaluation** checks into a **FastAPI** service, keeping agent responses accurate, safe and traceable across customer interactions.

AI Financial Analysis Chatbot | *LangChain, RAG, Embeddings, Python* MSc Project

- Launched a **RAG** chatbot over financial documents using **LangChain**, **embeddings** and **semantic search**, producing source-cited answers that support analyst research and decision-making.
- Validated retrieval quality using **chunking strategies**, **reranking** and **RAGAS** evaluation, reducing **hallucination** risk on numerical and regulatory financial questions.

μGenAI: Lightweight Generative AI Models | *PyTorch, Quantization, Distillation* MSc Project

- Compressed **transformer** models through **quantization** and **model distillation** to enable efficient **LLM** inference on low-memory edge hardware with minimal loss in output quality.
- Benchmarked compressed **LLMs** against full-precision baselines on latency, memory footprint and output quality to select the best **PyTorch** deployment configuration for resource-limited devices.

Candidrivo | *Python, LLM APIs, FastAPI* MSc Project

- Designed an **LLM**-powered application with a **FastAPI** backend, **prompt engineering** pipelines and **REST APIs**, taking it from concept to a working deployment.
- Implemented **Docker** containerisation and versioned prompt templates to keep the **LLM** workflow reproducible, testable and straightforward to deploy across environments.

Education

Nottingham Trent University United Kingdom
MSc Artificial Intelligence *Oct 2024 – Oct 2025*

Sandip Institute of Technology and Research Centre India
B.Eng. Mechanical Engineering *Jul 2021 – Jun 2024*

Certifications

Introduction to Retrieval Augmented Generation (RAG) Apr 2026
Coursera (Duke University) *ID: MVOFCAOMOKU3*

Generative AI with Large Language Models Apr 2026
Coursera (DeepLearning.AI and AWS) *ID: JDKM2NP85MG3*

MCP: Model Context Protocol May 2026
Coursera *ID: 82UBVOLJ9L49*

Quantitative Research Job Simulation Jun 2026
JPMorgan Chase and Co. x Forage *ID: ZDfxdomj4YYNARgab*

GenAI Powered Data Analytics Job Simulation Jun 2026
Tata Group x Forage *ID: TJBDXAPZsDMBg4NZB*

Microsoft Azure AI Fundamentals (AI-900) Sep 2025
Microsoft *ID: Wnm8J-2FMo*